Day 15 Revision Sheet - Tue, Nov 3, 2026

1. Key ideas

Excel - Pivot tables and charts

- Pivot table = summary without formulas. Four areas: Rows (groups), Columns (groups across), Values (the number + how: Sum, Average, Count, Max, Min), Filters (limit the whole table).
- Change Sum to Average: Value Field Settings > Summarize Values By. No new pivot needed.
- Pivot chart = chart tied to its pivot. Change the pivot, the chart follows. New source rows need Refresh first.
- Top seller by category: Value Filters > Top 10 > Top 1 by Sum of Sales. Max aggregation is the biggest single record, not the biggest category total.
- Source range: new rows outside it never appear, even after Refresh. Fix 1: Change Data Source and extend it. Fix 2: make the data an Excel Table (Ctrl+T) so it grows. A table expands the source; Refresh still has to be clicked.

Git - history, merge, clone, conflict

- Parents: first (root) commit 0, ordinary commit 1, merge commit 2 (the tips of both branches, not the root).
- Fast-forward: only the feature branch moved; main just slides its pointer, no new commit. Merge commit: BOTH branches have new commits (diverged). Git can still combine them; it only cannot fast-forward.
- git clone = copy a repository WITH history, sets origin, already initialised. git init = brand new empty repository. Uncommitted work on the other machine is not copied.
- Conflict = both branches changed the same lines of the same file differently. Markers: HEAD side = your current branch, after ======= = the branch being merged.
- Resolve: edit the file, remove all three marker lines, git add the file, git commit, git status shows working tree clean.

Verified by running in a real repo (not from memory):

```
git merge feature-x -> CONFLICT (content): Merge conflict in notes.txt, exit 1
edit file only, then: git commit -m "x"
  -> error: Committing is not possible because you have unmerged files.
  -> fatal: Exiting because of an unresolved conflict.   (exit 128)
git add notes.txt, git commit -> git status: nothing to commit, working tree clean
parents counted with git rev-list: root 0, C 1, B 1, merge commit 2
```

AI Fluency - prompting

- Clear instruction = what to do, format, length, audience. Few-shot = 2+ consistent INPUT -> OUTPUT pairs that show the pattern. Iteration = fix the output in the same chat, context is kept.
- Too-long output: fix the instruction first ("under 20 words"), change the examples only if it still fails. Good habit: ask the AI for a reusable prompt after a good result and save it.

Business - metrics (taught Nov 3)

- Metric: adding, averaging or counting it makes sense. Dimension: you group or filter by it (can be a number). Unit price and discount % are non-additive: use a weighted average.
- KPI = metric + goal: name, formula, period, target, owner. Conversion rate = orders / unique human visits x 100. Leading comes before the outcome and predicts it; lagging records it after. Always name the outcome first.

2. My mistakes and the fix

What I got wrong	Fix
Parents of A = 3; parents of D = none (called it the first commit)	A is the only root (0). D follows C so it has 1. Count the arrows going back.
Two parents of M: said 'feature-x commit and initial commit'	The two parents are B (main tip) and D (feature tip). A is only a shared ancestor.
Both branches have commits: 'Git cannot understand how to combine them'	Git CAN combine (merge commit). It cannot FAST-FORWARD. Wrong 5 times now: keep on watch list.
Few-shot examples were not input-output pairs (one was an instruction)	Each example = one Input line and its Output line, same pattern every time.
Prompt rewrite had no practical example and a vague audience	Say who it is for and ask for one example.
Source-range prevention: 'set a setting so it changes automatically'	Name it: Excel Table (Ctrl+T) or Change Data Source.
'Top-selling category' answered with Max of Total Sales	Value Filters > Top 10 > Top 1 by Sum of Sales.
Python retake: first-time step inside the pass condition	Two separate ifs: first-time step, then the update as its own if.

Day 15 Revision Sheet - page 2 of 2

Practice to retry, recall questions, open items

3. How I improved

- Pivot areas, Sum to Average, bar chart choice and the Table-vs-Refresh difference were all answered with reasons.
- Fast-forward: gave the correct reason (no new commit on main) when the two branches had not diverged.
- Ordered the conflict commands correctly (merge, edit, add, commit, status) without notes.
- Python: part b (topper, tie rule, sorted names) and part c (tuple slice, TypeError) were right on the retake.
- Prompting: spotted on my own that a good result should be turned into a saved reusable prompt.

4. Practice to retry (no solutions)

- Pivot: build it yourself in Excel. Data with Date, Category, Region, Sales (10+ rows). One pivot: Category in Rows, Region in Columns, Sum of Sales in Values. Group Date by month. Add a pivot chart. Add a row below the data and make the pivot show it.
- Pivot test: which two things can stop a new row from showing up? Name the exact menu path for one.
- Git graph: main has commits A,B,E; feature has C,D branching from A. Give the parent count of every commit after the merge, and say fast-forward or merge commit and why.
- Git hands-on, from a blank folder, no notes: init, notes.txt, branch, change the same line on both branches, merge, resolve, git status. Then clone your learning-log into a second folder and run git log --oneline.
- Prompting: for 'turn my daily notes into one-line learning logs', write 2 real Input/Output pairs and one line capping the length.
- Python: records of (student, subject, mark). Count passes per student, every student must appear. Test with the records in reverse order.

5. Recall questions (answer from memory)

- Name the four pivot areas and one job of each.
- A new row sits just below the pivot's source range. Does Refresh bring it in? What do you do?
- How many parents: first commit, normal commit, merge commit?
- When does Git fast-forward and when does it make a merge commit?
- What does a plain git commit do while conflicted files are still not added?
- What is the difference between git clone and git init?
- Few-shot vs clear instruction vs iteration: one line each.
- Metric or dimension, KPI parts, leading or lagging: define each in one line.
- Pass-count dictionary: key, value, first-time step, update.

6. Open items

- Business Practice 1-3 (KPIs, 10 fields, vanity metrics): submitted, feedback still to be given.
- Business case: 'Orders increased but profit decreased. Why?' - 5-line answer not written yet.
- AI Fluency part C (fix a 2000-word reply) is not in the record I have: redo it.
- Pivot and Git were revised as questions only. Hands-on build, group dates by month, clone to another folder and the conflict from a blank folder are still to do.
- Watch list: fast-forward vs merge commit; carrying the dict pattern into a new problem shape.
- Sat Nov 7: 15-minute unseen dict problem, no hints.